

CHAPTER 5: INTRODUCTION TO RISK, RETURN, AND THE HISTORICAL RECORD

1. The Fisher equation predicts that the nominal rate will equal the equilibrium real rate plus the expected inflation rate. Hence, if the inflation rate increases from 3% to 5% while there is no change in the real rate, then the nominal rate will increase by 2%. On the other hand, it is possible that an increase in the expected inflation rate would be accompanied by a change in the real rate of interest. While it is conceivable that the nominal interest rate could remain constant as the inflation rate increased, implying that the real rate decreased as inflation increased, this is not a likely scenario.
2. If we assume that the distribution of returns remains reasonably stable over the entire history, then a longer sample period (i.e., a larger sample) increases the precision of the estimate of the expected rate of return; this is a consequence of the fact that the standard error decreases as the sample size increases. However, if we assume that the mean of the distribution of returns is changing over time but we are not in a position to determine the nature of this change, then the expected return must be estimated from a more recent part of the historical period. In this scenario, we must determine how far back, historically, to go in selecting the relevant sample. Here, it is likely to be disadvantageous to use the entire data set back to 1880.
3.
$$1 + r_{\text{Real}} = \frac{1 + r_{\text{Nominal}}}{1 + i} = \frac{1 + .45}{1 + .30} = 1.1154 \rightarrow r_{\text{Real}} = 11.54\%$$
4. For the money market fund, your holding-period return for the next year depends on the level of 30-day interest rates each month when the fund rolls over maturing securities. The one-year savings deposit offers a 5% holding period return for the year. If you forecast that the rate on money market instruments will increase significantly above the current 3% yield, then the money market fund might result in a higher HPR than the savings deposit. The 20-year Treasury bond offers a yield to maturity of 5% per year, which is 100 basis points higher than the rate on the one-year savings deposit; however, you could earn a one-year HPR much less than 4% on the bond if long-term interest rates increase during the year. If Treasury bond yields rise above 5%, then the price of the bond will fall, and the resulting capital loss will wipe out some or all of the 5% return you would have earned if bond yields had remained unchanged over the course of the year.

5.
 - a. If businesses reduce their capital spending, then they are likely to decrease their demand for funds. This will shift the demand curve in Figure 5.1 to the left and reduce the equilibrium real rate of interest.
 - b. Increased household saving will shift the supply of funds curve to the right and cause real interest rates to fall.
 - c. Open market purchases of U.S. Treasury securities by the Federal Reserve Board are equivalent to an increase in the supply of funds (a shift of the supply curve to the right). The FED buys treasuries with cash from its own account or it issues certificates which trade like cash. As a result, there is an increase in the money supply, and the equilibrium real rate of interest will fall.
6.
 - a. The “Inflation-Plus” CD is the safer investment because it guarantees the purchasing power of the investment. Using the approximation that the real rate equals the nominal rate minus the inflation rate, the CD provides a real rate of 1.5% regardless of the inflation rate.
 - b. The expected return depends on the expected rate of inflation over the next year. If the expected rate of inflation is less than 3.5% then the conventional CD offers a higher real return than the inflation-plus CD; if the expected rate of inflation is greater than 3.5%, then the opposite is true.
 - c. If you expect the rate of inflation to be 3% over the next year, then the conventional CD offers you an expected real rate of return of 2%, which is 0.5% higher than the real rate on the inflation-protected CD. But unless you know that inflation will be 3% with certainty, the conventional CD is also riskier. The question of which is the better investment then depends on your attitude towards risk versus return. You might choose to diversify and invest part of your funds in each.
 - d. No. We cannot assume that the entire difference between the risk-free nominal rate (on conventional CDs) of 5% and the real risk-free rate (on inflation-protected CDs) of 1.5% is the expected rate of inflation. Part of the difference is probably a risk premium associated with the uncertainty surrounding the real rate of return on the conventional CDs. This implies that the expected rate of inflation is less than 3.5% per year.

7.

$$E(r) = [0.35 \times 44.5\%] + [0.30 \times 14.0\%] + [0.35 \times (-16.5\%)] = 14\%$$

$$\sigma^2 = [0.35 \times (44.5 - 14)^2] + [0.30 \times (14 - 14)^2] + [0.35 \times (-16.5 - 14)^2] = 651.175$$

$$\sigma = 25.52\%$$

The mean is unchanged, but the standard deviation has increased, as the probabilities of the high and low returns have increased.

8. Probability distribution of price and one-year holding period return for a 30-year U.S. Treasury bond (which will have 29 years to maturity at year-end):

Economy	Probability	YTM	Price	Capital Gain	Coupon Interest	HPR
Boom	0.20	11.0%	\$ 74.05	-\$25.95	\$8.00	-17.95%
Normal growth	0.50	8.0	100.00	0.00	8.00	8.00
Recession	0.30	7.0	112.28	12.28	8.00	20.28

9. $E(q) = (0 \times 0.25) + (1 \times 0.25) + (2 \times 0.50) = 1.25$

$$\sigma_q = [0.25 \times (0 - 1.25)^2 + 0.25 \times (1 - 1.25)^2 + 0.50 \times (2 - 1.25)^2]^{1/2} = 0.8292$$

10. (a) With probability 0.9544, the value of a normally distributed variable will fall within 2 standard deviations of the mean; that is, between -40% and 80%. Simply add and subtract 2 standard deviations to and from the mean.
11. From Table 5.5, the average risk premium Big/Value for the period 1927–2021 was: 12.02% per year.
Adding 12.02% to the 3% risk-free interest rate, the expected annual HPR for the Big/Value portfolio is: 3.00% + 12.02% = 15.02%.

12.

(01/1930-6/1974)						
	Small			Big		
	Low	2	High	Low	2	High
Average	0.99%	1.17%	1.48%	0.76%	0.81%	1.19%
SD	8.29%	8.38%	10.17%	5.70%	6.72%	8.89%
Skew	1.30	1.63	2.35	0.17	1.75	1.77
Kurtosis	9.74	13.10	17.69	7.06	17.80	14.64

(07/1974-12/2018)						
	Small			Big		
	Low	2	High	Low	2	High
Average	1.00%	1.35%	1.45%	0.99%	1.05%	1.13%
SD	6.69%	5.28%	5.49%	4.70%	4.35%	4.90%
Skew	-0.43	-0.55	-0.47	-0.33	-0.43	-0.54
Kurtosis	2.08	3.60	4.30	1.99	2.57	2.96

No. The distributions from (01/1930–06/1974) and (07/1974–12/2018) periods have distinct characteristics due to systematic shocks to the economy and subsequent government intervention. While the returns from the two periods do not

differ greatly, their respective distributions tell a different story. The standard deviation for all six portfolios is larger in the first period. Skew is also positive, but negative in the second, showing a greater likelihood of higher-than-normal returns in the right tail. Kurtosis is also markedly larger in the first period.

$$13. \quad a. \quad r_{\text{real}} = \frac{1 + r_{\text{nominal}}}{1 + i} - 1 = \frac{r_{\text{nominal}} - i}{1 + i} = \frac{0.80 - 0.70}{1.70} = 0.0588, \text{ or } 5.88\%$$

$$b. \quad r_{\text{nominal}} - i = .80 - .70 = .10 \approx r_{\text{real}}$$

Clearly, the approximation gives a real HPR that is too high.

CFA PROBLEMS

1. The expected dollar return on the investment in equities is \$18,000 ($0.6 \times \$50,000 + 0.4 \times -\$30,000$) compared to the \$5,000 expected return for T-bills. Therefore, the expected risk premium is \$13,000.
2. $E(r) = [0.2 \times (-25\%)] + [0.3 \times 10\%] + [0.5 \times 24\%] = 10\%$
3. $E(r_X) = [0.2 \times (-20\%)] + [0.5 \times 18\%] + [0.3 \times 50\%] = 20\%$
 $E(r_Y) = [0.2 \times (-15\%)] + [0.5 \times 20\%] + [0.3 \times 10\%] = 10\%$
4. $\sigma_X^2 = [0.2 \times (-20 - 20)^2] + [0.5 \times (18 - 20)^2] + [0.3 \times (50 - 20)^2] = 592$
 $\sigma_X = 24.33\%$
 $\sigma_Y^2 = [0.2 \times (-15 - 10)^2] + [0.5 \times (20 - 10)^2] + [0.3 \times (10 - 10)^2] = 175$
 $\sigma_Y = 13.23\%$
5. $E(r) = (0.9 \times 20\%) + (0.1 \times 10\%) = 19\% \rightarrow \$1,900$ in returns
6. The probability that the economy will be neutral is 0.50, or 50%. Given a neutral economy, the stock will experience poor performance 30% of the time. The probability of both poor stock performance and a neutral economy is therefore:
 $0.30 \times 0.50 = 0.15 = 15\%$
7. $E(r) = (0.1 \times 15\%) + (0.6 \times 13\%) + (0.3 \times 7\%) = 11.4\%$